

Functional Requirements Document (FRD)

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Project Name:- MindForge AI - Personalized Learning App

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1. Introduction

1.1 Purpose

This Functional Requirements Document (FRD) defines the functional behavior of the MindForge AI platform based on the approved business requirements documented in the Business Requirements Document (BRD). It describes how each system feature should operate, including user interactions, business rules, validations, and expected outputs.

This document serves as a reference for developers, QA engineers, and other project stakeholders during the design, implementation, and testing phases.

1.2 Scope

The FRD covers all major functional modules included in the first release of MindForge AI, including:

- User Authentication
- SkillScan AI
- Personalized Learning Roadmap
- Learn & Quiz
- Visual Concept Cards

- Knowledge Decay Tracker
 - Concept Confusion Detector
 - Progress Dashboard
 - PDF Generation
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2. Functional Overview

The platform consists of the following functional modules.

Module	Description
Authentication	Manages user registration, login, and profile access.
SkillScan AI	Assesses the learner's current knowledge and learning objectives.
Roadmap Generator	Creates personalized learning roadmaps based on SkillScan results.
Learn & Quiz	Delivers AI-generated learning content and quizzes.
Visual Concept Cards	Generates concise revision cards for key concepts.
Knowledge Decay Tracker	Tracks retention and recommends revision schedules.
Concept Confusion Detector	Identifies topics where the learner shows repeated misunderstandings.
Progress Dashboard	Displays learner progress, analytics, and achievements.
PDF Generator	Exports AI-generated learning material as downloadable PDFs.

3. Functional Requirements

Each functional requirement traces back to a corresponding Business Requirement (BR).

FR-001 – User Registration & Authentication

Field	Description
Requirement ID	FR-001
Related BR	BR-001
Module	Authentication
Priority	High
Actor	Learner
Description	The system shall allow new users to create an account and authenticate using their registered credentials.
Trigger	User selects "Sign Up" or "Login".
Preconditions	User has access to the application.
Postconditions	User account is created or authenticated successfully.

Inputs

- Full Name
- Email Address
- Password

Outputs

- Successful registration confirmation
- Successful login and dashboard access

Business Rules

- Email address must be unique.
- Password must meet defined security requirements.
- Authenticated users are redirected to the dashboard.

Validation Rules

- Mandatory fields cannot be empty.

- Email must follow a valid format.
- Password must contain the minimum required characters.

Exception Handling

- Duplicate email
- Invalid credentials
- Server unavailable

FR-002 – SkillScan AI

Field	Description
Requirement ID	FR-002
Related BR	BR-002
Module	SkillScan AI
Priority	High
Actor	Learner
Description	The system shall assess the learner's current knowledge level and identify learning objectives using AI.
Trigger	User initiates the SkillScan assessment.
Preconditions	User is logged in.
Postconditions	Assessment results are generated and stored.

Inputs

- Learning Goal
- Current Skill Level
- Preferred Learning Domain
- User Responses

Outputs

- Skill Assessment Report
- Learning Level
- Recommended Learning Focus Areas

Business Rules

- Assessment must be completed before generating a roadmap.

- Results are stored for future reference.
- Users may retake the assessment.

Validation Rules

- Learning goal is mandatory.
- Assessment questions must be completed.

Exception Handling

- AI response timeout
- Incomplete assessment
- API failure

FR-003 – Personalized Learning Roadmap

Field	Description
Requirement ID	FR-003
Related BR	BR-003
Module	Roadmap Generator
Priority	High
Actor	Learner
Description	The system shall generate a personalized learning roadmap using the SkillScan results and learner goals.

Inputs

- Skill Assessment Results
- Learning Goal
- Time Availability

Outputs

- Personalized Learning Roadmap
- Estimated Completion Timeline

Business Rules

- Roadmap generation requires a completed SkillScan.
- Roadmaps can be regenerated after reassessment.
- Learning milestones must be displayed sequentially.

FR-004 – Learn & Quiz

The system shall:

- Recommend AI-generated learning resources.
- Generate quizzes after topic completion.
- Calculate quiz scores.
- Store learner performance.
- Recommend revision based on quiz performance.

FR-005 – Visual Concept Cards

The system shall:

- Generate concise AI-powered revision cards.
- Highlight key definitions, formulas, and concepts.
- Allow learners to review cards multiple times.

FR-006 – Knowledge Decay Tracker

The system shall:

- Track completed learning activities.
- Estimate knowledge retention over time.
- Notify learners when revision is recommended.
- Update retention metrics after revision.

FR-007 – Concept Confusion Detector

The system shall:

- Analyze quiz responses and learner interactions.
- Detect frequently misunderstood concepts.
- Recommend targeted revision resources.

FR-008 – Progress Dashboard

The dashboard shall display:

- Learning Progress
- Quiz Performance
- Roadmap Completion
- Knowledge Retention Status
- Recommended Next Activities

FR-009 – PDF Generation

The system shall:

- Generate downloadable PDF study material.
- Include summaries, concept cards, and quizzes.
- Allow users to download generated documents.

4. Business Rules

Rule ID	Business Rule
BRL-001	Users must complete SkillScan before roadmap generation.
BRL-002	Quiz performance updates learner progress metrics.
BRL-003	Knowledge retention is recalculated after each revision session.
BRL-004	Concept confusion recommendations are based on historical learner performance.
BRL-005	PDF generation is available only after learning content has been generated.

5. Validation Rules & Error Handling

Scenario	Expected System Behavior
Invalid Login	Display appropriate error message.
Missing Required Fields	Highlight mandatory fields.
AI Service Failure	Inform the user and allow retry.
Empty Quiz Submission	Prevent submission until answered.
PDF Generation Failure	Display error and provide retry option.

6. Acceptance Criteria

Functional Requirement	Acceptance Criteria
FR-001	Users can successfully register and log in.
FR-002	SkillScan generates a learner assessment.
FR-003	Personalized roadmap is generated based on assessment.
FR-004	Quizzes are generated and scored correctly.
FR-005	Visual Concept Cards are created for completed topics.
FR-006	Knowledge Decay Tracker recommends revisions appropriately.
FR-007	Concept Confusion Detector identifies weak concepts.
FR-008	Dashboard displays accurate learner progress.
FR-009	PDF export generates downloadable study material.

7. Requirement Traceability

Business Requirement	Functional Requirement
BR-001	FR-001
BR-002	FR-002
BR-003	FR-003
BR-004	FR-004
BR-005	FR-005
BR-006	FR-006
BR-007	FR-007
BR-008	FR-008
BR-009	FR-009
BR-010	FR-010

8. Appendices

Glossary

Term	Description
BR	Business Requirement
FR	Functional Requirement
LLM	Large Language Model
API	Application Programming Interface
UI	User Interface
AI	Artificial Intelligence
